

Internet Security Flow Analysis

Understanding Internet Communication Through a Cybersecurity Perspective

Technical Project Report

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Areas of Focus

- Security Operations (SOC)
- Microsoft Azure
- Cloud Security
- Networking Fundamentals
- Windows Security
- Linux Fundamentals

Project Version

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Project Repository

GitHub

<https://github.com/Siambabon/internet-security-analysis>

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Professional Statement

"Cybersecurity begins with understanding how systems communicate, where trust exists, and how defensive controls protect modern digital environments."

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This project was developed for educational and professional portfolio purposes.

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INTRODUCTION & PROJECT BACKGROUND

1. Introduction

The **internet powers** almost every part of our daily lives, from banking and healthcare to education and communication. While using a website feels like a simple action, it actually relies on many complex systems working together behind the scenes. Because we rely on these systems so heavily, **protecting them from cyber threats** is essential.

To defend against attacks effectively, **cybersecurity professionals** must first understand how information moves. You cannot secure what you don't understand; therefore, knowing how technology talks to technology is the **foundation of any strong security strategy**.

This project explores that foundation by following the **full 'journey' of an internet request**—from the moment you type a website address to the moment the page loads. By **combining networking and security** into a single workflow, this report maps out where communication happens, where trust is placed, and where security risks can appear.

Throughout this project, I have analyzed each step from a 'defender's' perspective. I identify common vulnerabilities and pair them with practical security controls to demonstrate how to stop potential threats. By connecting networking concepts with defensive principles, this project encourages the analytical thinking needed to protect digital environments.

This report serves as a foundational cybersecurity analysis, demonstrating the structured thinking and technical knowledge required for the field. It is a key part of my professional portfolio, highlighting my focus on **Security Operations (SOC)**, **Azure, Cloud Security**, and **modern defensive strategies**.

2. Project Objectives & Scope

Project Purpose

The primary purpose of this project is to develop a **practical understanding** of how internet communication functions from a cybersecurity perspective. Rather than focusing solely on networking concepts or individual security technologies, this project examines the **complete communication process** to understand how trust is established between systems, where vulnerabilities may arise, and how **defensive controls** help reduce security risks.

This project was designed as a learning-oriented technical analysis that **bridges the gap** between **networking fundamentals** and **modern defensive cybersecurity practices**. It emphasizes analytical thinking, security awareness, and the practical application of foundational cybersecurity concepts.

Project Objectives

The objectives of this project are to:

- **Lifecycle of Internet Communication:** Develop a structured understanding.
 - **Attack Surfaces:** Identify common points of vulnerability throughout the process.
 - Analyze how attackers may exploit weaknesses at different stages.
 - Examine the role of **defensive security controls** in protecting digital systems.
 - Apply **fundamental cybersecurity principles** to real-world communication scenarios.
 - Strengthen **analytical thinking** from both an attacker's and a defender's perspective.
 - Build a strong technical foundation for future learning in **Security Operations (SOC)**, **Cloud Security**, and **Microsoft Azure**.
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Project Scope

This project focuses on the **defensive analysis** of internet communication and does **not** involve offensive security techniques or unauthorized system access.

The scope includes:

- **Internet communication workflow**
- **DNS resolution**
- **Client–server communication**
- **Common cyber threats**
- **Defensive security controls**
- **Security principles**
- **Risk awareness**
- **Foundational cybersecurity concepts**

The project is intended for educational purposes and serves as part of a professional cybersecurity portfolio.

3. Understanding Internet Communication

Every interaction on the Internet begins with a simple action performed by a user. Whether accessing a website, checking email, or using a cloud application, the request travels through multiple **interconnected systems** before reaching its destination. Each component within this process performs a specific function, and together they enable **secure and reliable communication** across the Internet.

Although this communication often appears instantaneous, it involves several technologies working together behind the scenes. Domain Name System (DNS) services translate human-readable domain names into IP addresses; browsers generate requests using standard communication protocols; routers forward packets across networks; firewalls inspect and filter traffic; and web servers process incoming requests before returning the appropriate response.

From a cybersecurity perspective, every stage introduces a **trust relationship** between systems. Whenever one system trusts another, there is an opportunity for attackers to **exploit that trust** through techniques such as spoofing, interception, unauthorized access, or malicious input. **Understanding these trust relationships** is essential because **cybersecurity is not only about preventing attacks**—it is also about **understanding how systems communicate** so that **security controls** can be applied effectively.

A strong understanding of internet communication provides the foundation required for Security Operations Centre (SOC), incident response, cloud security, vulnerability management, and many other cybersecurity disciplines. For this reason, **networking fundamentals** remain one of the most important building blocks for every **cybersecurity professional**.

4. Internet Communication Workflow

The following workflow illustrates the typical path followed when a user accesses a website over the Internet.



Each stage within this workflow performs a unique function while introducing different security considerations.

****User****

The communication process begins when a user enters a website address or submits a request through a web browser. User behaviour plays a significant role in cybersecurity because **human error** remains one of the most common causes of security incidents.

****DNS Resolution****

The Domain Name System translates the requested domain name into an IP address so that the browser can locate the correct server. Since DNS establishes trust between the user and the destination, it is frequently targeted by attackers through techniques such as **DNS spoofing** and **cache poisoning**.

****Web Browser****

The browser creates an HTTP or HTTPS request and initiates communication with the destination server. Modern browsers also perform **certificate validation** and enforce various security policies that help protect users during web communication.

****Internet****

The request travels through multiple routers, switches, and service providers before reaching its destination. During this stage, data may become vulnerable if **secure communication protocols** are not properly implemented.

****Firewall****

Firewalls inspect incoming and outgoing traffic according to predefined security rules. Their primary purpose is to reduce **unauthorized access** while allowing legitimate communication to continue.

****Web Server****

The server receives the client's request, processes it, and forwards the necessary information to the application layer. Secure server configuration is critical for reducing the risk of **unauthorized access** and system compromise.

****Application & Database****

The application processes business logic and interacts with the database to retrieve or store information. Secure coding practices and **input validation** are essential to prevent attacks such as **SQL Injection** and unauthorized data access.

****Response****

Finally, the server returns the requested information to the user. Throughout this process, **encryption**, **authentication**, and **integrity verification** help ensure that the response remains secure and trustworthy during transmission.

5. Project Architecture Analysis

Understanding the Communication Flow

The architecture presented in this project illustrates the **journey of a typical web request** and highlights how **multiple technologies interact** to **deliver information securely** between a user and a web application. Although this process usually completes within milliseconds, every stage introduces **trust relationships**, **communication protocols**, and **potential security risks** that must be carefully managed.

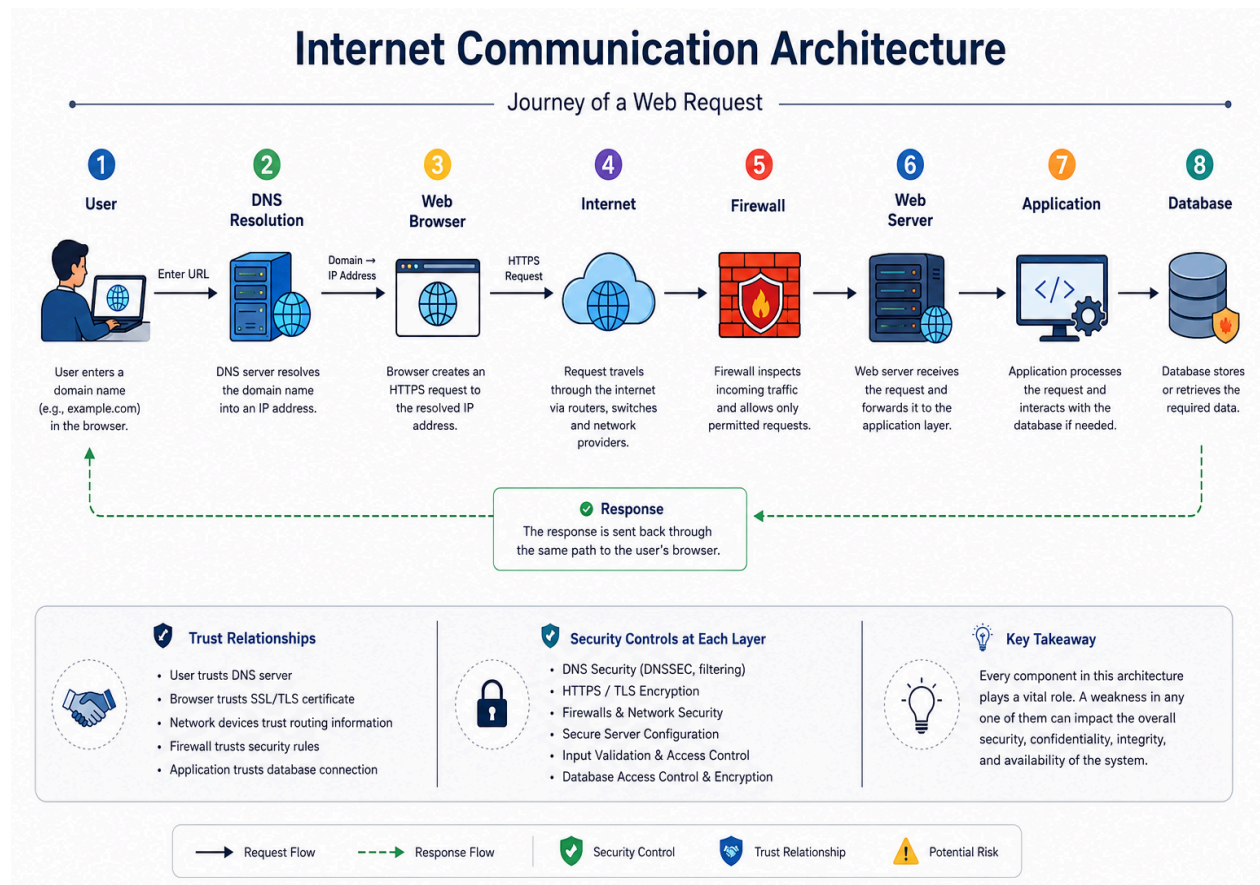


Figure 1. Internet communication workflow illustrating the relationship between the user, networking components, security controls, application services, and data storage.

The communication process begins when a user requests access to a website by entering a domain name into a web browser. The browser relies on the **Domain Name System (DNS)** to resolve the domain into an IP address, allowing the request to locate the correct destination server.

After DNS resolution, the browser generates an **HTTP or HTTPS request** and sends it across multiple network devices before reaching the destination environment. During transmission, routers, switches, and Internet Service Providers forward packets while **firewalls inspect traffic** according to predefined security policies.

Once the request reaches the **web server**, it is processed by the **application layer**. If required, the application retrieves or stores information within a **database** before

generating a response that travels back through the same communication path to the user's browser.

Although the workflow appears linear, every component depends on trust. A compromised DNS server, an exposed web application, or an improperly configured firewall can significantly affect the **confidentiality, integrity, and availability** of information throughout the communication process.

From a cybersecurity perspective, understanding these relationships is essential because effective security depends on **protecting every stage of communication** rather than focusing on a single technology.

6. Attack Surface Analysis

Understanding Potential Threats

Every connected system exposes some level of **attack surface**. An **attack surface** represents any point where an attacker may attempt to interact with, exploit, or compromise a system. The objective of defensive cybersecurity is not to eliminate every possible risk—an impossible task—but to reduce the attack surface through **secure design, proper configuration, continuous monitoring, and layered security controls**.

The following table summarises common attack opportunities identified throughout the communication workflow.

Communication Stage	Potential Threat	Security Impact	Risk Level
DNS Resolution	DNS Spoofing	Redirects users to malicious destinations	High
Network Communication	Man-in-the-Middle (MITM)	Data interception or modification	High
Firewall Configuration	Misconfigured Rules	Unauthorized network access	Medium
Web Server	Service Misconfiguration	System compromise	High
Application Layer	Injection Attacks	Data manipulation and unauthorized actions	High
Database	Unauthorized Access	Exposure of sensitive information	High
User	Phishing & Social Engineering	Credential theft	High

Security Perspective

One important lesson from this analysis is that attackers rarely target only one technology. Instead, they attempt to identify the **weakest point** within the communication process. In many cases, **human behaviour**, **insecure configurations**, or outdated software provide easier opportunities than sophisticated technical exploits.

For this reason, modern cybersecurity emphasises **Defence in Depth**, where multiple independent security controls work together to reduce the likelihood and impact of successful attacks.

Attack surface analysis also supports **risk-based decision-making** by helping organisations prioritise security improvements according to business impact, likelihood of exploitation, and available defensive capabilities.

7. Defensive Security Controls

Cybersecurity is fundamentally concerned with **reducing risk** rather than eliminating it. Organisations implement multiple defensive controls to protect systems, detect suspicious activity, and maintain the **confidentiality, integrity, and availability** of information.

The following controls play a critical role throughout the Internet communication workflow.

Security Control	Primary Purpose
Firewall	Filters and controls network traffic according to security policies.
HTTPS / TLS	Encrypts communication between clients and servers to protect data in transit.
Multi-Factor Authentication (MFA)	Adds an additional layer of identity verification beyond passwords.
Secure Configuration	Reduces unnecessary exposure by disabling insecure services and applying security best practices.
Input Validation	Prevents malicious user input from affecting application behaviour.
Monitoring & Logging	Records system activity and supports incident detection and forensic investigations.
Virtual Private Network (VPN)	Protects remote communication through encrypted network tunnels.

Defense in Depth

Modern organizations recognize that no single security control can prevent every cyber attack. For this reason, they implement **Defence in Depth**—a **layered security strategy** where multiple independent controls work together to reduce risk. If one control fails, additional layers continue to protect the environment. For example, if a phishing attack compromises a user's password, **Multi-Factor Authentication (MFA)** can still prevent unauthorized access. Similarly, **HTTPS** encrypts sensitive information during transmission, reducing the risk of interception across untrusted networks. By combining **preventive, detective, and corrective controls**, Defence in Depth improves an organization's overall **security posture** and **operational resilience**.

Key Takeaways

- The analysis presented in this project demonstrates that effective **cybersecurity** depends on implementing multiple layers of protection throughout the communication process rather than relying on a single security control.
- Each defensive measure contributes to reducing organizational risk by preventing attacks, detecting suspicious activities, limiting the impact of security incidents, and supporting rapid recovery when security events occur.
- Rather than functioning independently, these controls complement one another to create a stronger and more resilient security posture. Together, they help protect users, applications, network infrastructure, and sensitive information against a wide range of evolving cyber threats.
- Understanding how these defensive controls interact throughout the communication workflow provides an essential foundation for modern cybersecurity practices. It reinforces the importance of adopting a **layered security** strategy to ensure **confidentiality, integrity, and availability**.

Conclusion

Defensive security controls form the foundation of modern cybersecurity. While no individual security mechanism can completely eliminate cyber threats, implementing multiple complementary controls significantly reduces organizational risk and strengthens overall resilience.

Throughout this project, the communication workflow demonstrated how technologies such as firewalls, HTTPS/TLS, Multi-Factor Authentication (MFA), secure configuration, monitoring and logging, input validation, and Virtual Private Networks (VPNs) work together to protect digital systems at different stages of communication.

By applying the principle of **Defence in Depth**, organizations improve their ability to prevent attacks, detect malicious activity, respond effectively to security incidents, and maintain the confidentiality, integrity, and availability of critical information.

Developing a clear understanding of these defensive security controls is an essential step toward building practical skills in **Security Operations (SOC)**, **Cloud Security**, **Microsoft Azure**, Incident Response, and other cybersecurity disciplines. As cyber threats continue to evolve, implementing layered security remains one of the most effective approaches for protecting modern digital environments.